David C. Morley

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Applied Research Scientist — Speech, NLU & Clinical AI

I build voice AI that puts humans first: from aircraft maintainers and surgeons to children learning to speak.

EXPERIENCE

Research Scientist, Cobalt Speech & Language — Remote Aug 2022 – Present

Clinical speech & language AI (2025 – present)

- Designed and trained the neural models behind SALT's automated transcript coding (segmentation, mazes, punctuation, error codes, code-switching), launching with paying clinicians at the Nov 2026 ASHA convention. On human transcripts: segmentation F1 0.40 → 0.94 and MLU agreement with human coders ICC 0.02 → 0.98; error coding, built from scratch, reaches ICC 0.90 on each child's error rate.
- Rebuilt evaluation around one canonical scorer, fixing defects that had understated model quality (same model: F1 0.61 → 0.90) and a normative-scoring bug that inflated impairment scores, both before they reached clinicians.
- Built data-provenance and license guards that check every training record against its permissions and stamp each model with the data behind it; the approach is being adopted on other projects.
- Raised children's pronunciation-error detection from 79.9% to 86.8% with fewer false flags. Prototyped free-speech scoring that ignores adult speakers for an ASHA 2026 convention demo.
- Built a dashboard showing how transcript errors shift clinical measures, with an exact small-cell privacy audit.

Operating-room voice control · technical lead (2024 – present)

- Project lead for a surgical-device customer's voice-control product: scope statements of work, run customer technical calls, and own every release since fall 2024.
- Replaced the SVM intent classifier with distilled BERT (unseen-wording accuracy 50% → 84%); a calibrated reject class and hard negatives more than halved false activations (22.5% → 10%) at 98.3% task completion.
- Built the end-to-end task-completion harness for releases; grew regression tests from 35 audio clips to 600+.

U.S. Air Force SBIR programs (2022 – 2025)

- Sole NLP specialist on a 3-person team building a hands-free voice assistant for aircraft maintainers. Built the domain language model and lexicon, fine-tuned the acoustic model on flight-line noise, and built the NLU, the dialog, and a parts-and-procedures knowledge graph with fuzzy voice search.
- Led GUI and voice-UX design for the Phase II inspection-and-reporting app and built a form-to-dialog-model compiler. Primary demoer, from on-site technician sessions to briefings for colonels.

Engineering practice

- Built AI-assisted code-review tooling (custom Claude Code skills and agents) that writes mutation-tested regression tests and finds bugs. Reviewed 430+ teammates' pull requests since 2022. Mentored a summer research intern.

Research Intern, Pryon — Boston, MA Jan 2022 – May 2022

- Built a model that detects follow-up questions and restores their elided context for a customer-facing QA product.

Linguistic Data Intern, Basis Technology — Somerville, MA Jun 2021 – Aug 2021

- Built a model to detect structured text regions that beat the existing approach, and presented it to stakeholders.

SKILLS

- **ML & NLP:** PyTorch, Hugging Face Transformers (BERT, DeBERTa, Qwen, T5), CRFs, ONNX Runtime, LoRA
- **Speech:** ASR customization (LMs, lexicons, grammars, acoustic models), Whisper, wav2vec, NeMo
- **Evaluation:** test-set design, bootstrap confidence intervals, ICC/CCC agreement
- **Software:** Python, Go, JavaScript, SQL, Vue, React, Nix, Docker, GitHub Actions, GCP
- **Linguistics:** IPA/X-SAMPA transcription, language-sample analysis, dialog design

EDUCATION

M.S., Computational Linguistics — Brandeis University 2022

B.S., Applied & Computational Mathematics — Brigham Young University 2020

ADDITIONAL

- **Languages:** Kiribati (conversational; former interpreter), Spanish (intermediate). **Side project:** hand-built multiplayer drawing game (Go/PocketBase, Vue; race and E2E tests), github.com/morleyd